

Kausha Trivedi

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EDUCATION

Northeastern University , Boston, MA	Jan 2023 - Apr 2025
Master of Science in Computer Science	GPA: 3.8/4.0
Indus University , Ahmedabad, Gujarat	Sept 2018 - May 2022
Bachelor of Science in Computer Science	GPA: 4.0/4.0

WORK EXPERIENCE

GBCS Group , Boston, MA	Jun 2024 - Dec 2024
Software Engineer Intern	
- Spearheaded the development of backend services and RESTful APIs using Python, processing over 100,000 daily events with sub-200ms latency by optimizing PostgreSQL queries and implementing a Redis cache. - Shipped a real-time analytics dashboard in React and TypeScript, providing stakeholders with a critical tool that visualized complex data streams and improved business decision-making. - Authored a shared component library that accelerated feature development by 30%, establishing a framework that enabled the team to rapidly build and iterate on data-intensive UIs. - Automated the CI/CD pipeline using AWS and Docker, enabling the team to safely and frequently ship features for a mission-critical financial product. - Increased application reliability by implementing a comprehensive testing suite with PyTest, achieving 85% code coverage for critical backend services and preventing regressions.	
Elsner Technology , Ahmedabad, Gujarat	May 2021 - Nov 2022
Software Developer	
- Commanded the full-stack development for a high-traffic marketplace, designing the SQL database schema and engineering the APIs and React frontend to support 50,000+ daily active users. - Led the migration of a monolithic payment module to an independent microservice, improving system reliability, fault isolation, and enabling dedicated scaling for a revenue-critical component. - Launched a full-stack monitoring platform that tracked the health of 100+ microservices, reducing the mean time to detect critical production incidents by 60%. - Slashed database query times by 50% on key read operations by redesigning table schemas, adding composite indexes, and refactoring complex joins in PostgreSQL. - Decreased initial page load time by 60% (from ~4.5s to 1.8s) by implementing code splitting in Webpack, lazy-loading non-critical assets, and integrating a CDN for static content. - Hardened the security of the payment processing module by implementing defenses against common vulnerabilities (XSS, CSRF) and integrating secure, token-based authentication flows.	

PROJECTS

AI-Powered SaaS Platform <i>Python, FastAPI, TypeScript, React, OpenAI GPT-4, PostgreSQL</i>	
- Architected an end-to-end, AI-powered application from concept to deployment, directly integrating the OpenAI GPT-4 API to solve a real-world business problem. - Constructed the complete system, from the Python/FastAPI backend and SQL database to the human-centric React/TypeScript user interface, demonstrating full-stack ownership. - Pioneered a powerful abstraction that automated 85% of user inquiries, showcasing the practical application of modern LLMs to enhance productivity and user experience.	
Internal Data Visualization & Tooling Platform <i>React, TypeScript, GraphQL</i>	
- Forged a centralized dashboard to process and visualize 10GB of complex data per session, empowering internal teams with a tool that achieved sub-500ms load times. - Crafted a library of 15+ reusable visualization components, creating a streamlined system that enabled other engineers to rapidly build and deploy new internal data tools. - Championed the adoption of GraphQL, designing a schema that eliminated over-fetching from traditional REST APIs and enabled the frontend to request data with surgical precision.	

TECHNICAL SKILLS

Languages & Frameworks:	Python, Java (Spring Boot), Node.js, JavaScript/TypeScript, C++, SQL, PHP
Frontend:	React, Next.js, Vue.js, Component Libraries, UI/UX, Performance Optimization
Databases & Cache:	PostgreSQL, MongoDB, Redis, Query Optimization, Database Scaling Strategies
Backend & APIs:	Microservices, REST, WebSocket, GraphQL, Express.js, JWT/RBAC, Laravel
Infrastructure & DevOps:	AWS (ECS, Lambda), Docker, Jenkins, CI/CD, ELK Stack, OpenTelemetry
AI & Data:	LLMs (OpenAI GPT-4), Data Modeling, Data Pipelines, FastAPI